

# Interactive Medical Image Diagnosis with Chatbot Assistance

Essam A. Rashed, Ahmed T. Elboardy, Yiming Jia, Mina Yousef

*Graduate School of Information Science,*

*University of Hyogo,*

Kobe, Japan

rashed@gsis.u-hyogo.ac.jp, {af25s001,af26l003,af26v009}@guh.u-hyogo.ac.jp

Ziad Elshaer, Ghada Khoriba

*School of Inf. Tech. and Comp. Science,*

*Nile University,*

Giza, Egypt

{aelshaer,ghadakhoriba}@nu.edu.eg

**Abstract**—This paper presents an ongoing research project funded by JST PRESTO that aims to develop an interactive medical image diagnosis system that integrates multimodal deep learning and conversational AI. The project focuses on enabling automated radiology report generation, clinical decision support, and human-centered interaction through a unified framework. We describe the project objectives, structure, current progress and preliminary findings, and highlight opportunities for collaboration and future development.

**Index Terms**—Research Project, Medical Imaging, Chatbot, Multimodal AI, Clinical Decision Support

## I. PROJECT OVERVIEW

**Project Title:** Interactive Medical Image Diagnosis with Chatbot Assistance

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**Funding:** JST PRESTO, Japan (Grant No. JPMJPR23P7)

**Principal Investigator:** E. A. Rashed (presenter)

**Research Assistants:** A. T. Elboardy, Y. Jia, M. Yousef

**Institution:** University of Hyogo, Japan

**Collaborators:** Z. Elshaer and G. Khoriba

**More information:** available online [1]

## II. INTRODUCTION

Medical imaging has become a fundamental component of modern healthcare systems, supporting a wide range of applications including disease diagnosis, treatment planning, and longitudinal monitoring [2]. Despite significant advances in imaging technologies and computational methods, the increasing volume and complexity of medical data have introduced substantial challenges for clinicians, particularly radiologists who are required to interpret large numbers of images under time constraints [3]. The emergence of deep learning has enabled remarkable progress in tasks such as image segmentation, classification, and anomaly detection, achieving performance levels that approach or even exceed human expertise in certain domains [4]. However, these advances are often limited to isolated tasks and lack integration into comprehensive clinical workflows.

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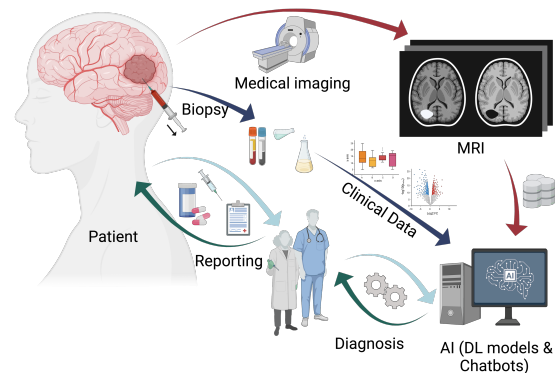


Fig. 1. Overview of the research project framework with the clinical application of brain cancer diagnosis using MRI.

In parallel, large language models (LLMs) have demonstrated unprecedented capabilities in natural language understanding and generation, enabling applications such as automated report writing, question answering, and conversational assistance [5]. Recent developments in vision-language models (VLMs) have further enabled the joint processing of visual and textual data, offering new opportunities for radiology report generation and clinical reasoning [6]. Nevertheless, most existing approaches focus on static report generation and do not address the interactive and dynamic nature of clinical decision-making. Furthermore, concerns related to reliability, explainability, and ethical deployment remain critical barriers to adoption in healthcare settings.

This research project proposes an interactive medical image diagnosis framework that integrates deep learning-based visual analysis for medical image diagnosis. The goal is to create a unified system that not only performs accurate image-based diagnosis but also facilitates meaningful interaction with users, including clinicians, trainees, and patients. By combining multimodal AI with interactive reasoning, the proposed framework aims to bridge the gap between algorithmic performance and practical clinical usability. An overview illustration of this project is shown in Fig. 1.

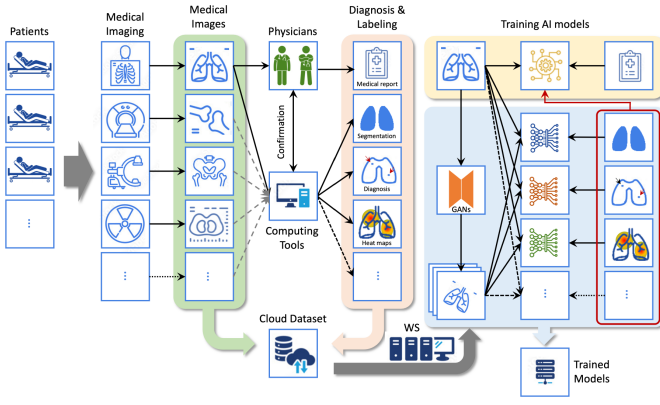


Fig. 2. Schematic illustration of data flow of image analysis module.

### III. OBJECTIVES AND EXPECTED OUTPUTS

This research project addresses key challenges in medical AI deployment, including data complexity, lack of explainability, and limited interaction with users. The project aims to develop a unified diagnostic system that integrates image analysis with conversational reasoning. This framework enable fast, accurate and affordable pipeline for medical image diagnosis, generation of diagnosis report, communication with patient inquiries, and related education purposes for junior medical staff. The main research stream can be split into two tracks.

- The development of multi-task image diagnosis models using several deep learning architectures. This process will lead to extract extensive image-based features for accurate diagnosis and abnormality detection.
- A chatbot-enabled medical report generator that can imitate the performance of expert radiologist in producing text-based medical reports.

All developed models will be merged in a general integrated framework for potential clinical use. This project is expected to improve the healthcare sector in Japan (and worldwide as well) through formulating innovative framework considering of emerging technology of generative AI especially with the expected limitation of healthcare manpower in the near future. The project objectives include:

- Develop multimodal AI for medical image understanding
- Enable chatbot-based interaction for clinical reasoning
- Incorporate human-in-the-loop validation

The outcomes of this research project are expected to contribute to:

- Interactive medical diagnosis system. We mainly focus on the development of teleradiology system for diagnosis of brain cancer from MRI scans.
- Validation of the proposed framework using real clinical workflow
- Research publication and open-source tools.

### IV. RELEVANCE TO COMPSAC

This research project is strongly aligned with the core themes and interdisciplinary vision of IEEE COMPSAC. It

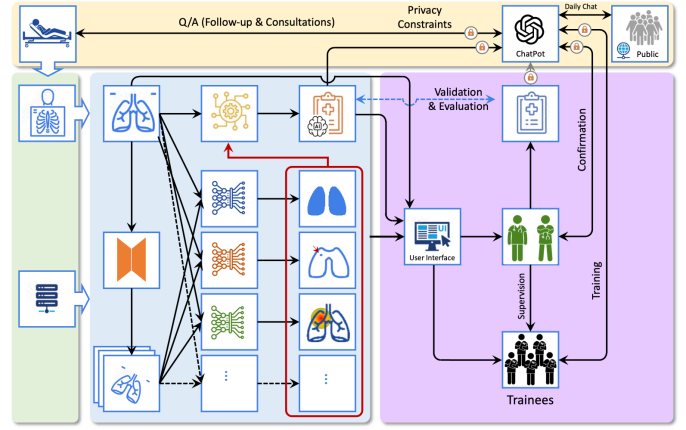


Fig. 3. Integrated framework for chatbot-enabled medical image diagnosis.

advances intelligent systems by integrating multimodal AI, contributes to healthcare computing through practical clinical applications, and reflects modern software system design principles including modularity and scalability. The project also emphasizes trustworthiness through explainability and human-in-the-loop validation, while combining multiple disciplines such as computer vision, NLP, and HCI. This alignment highlights its contribution to both theoretical and applied aspects of computing research within COMPSAC.

### V. PROJECT ORGANIZATION

The proposed framework is designed as an end-to-end system that integrates multimodal data processing, knowledge extraction, and interactive communication. The architecture consists of two main components: a deep learning-based medical image analysis module and a chatbot-assisted report generation and interaction module. These components are connected through a structured representation of extracted clinical information, enabling seamless communication between visual and textual processing stages.

The image analysis module is responsible for extracting clinically relevant information from medical images (Fig. 2). This includes tasks such as abnormality segmentation, organ localization, disease classification, and characterization. Advanced deep learning architectures are employed to capture both local and global features, leveraging convolutional and transformer-based mechanisms to model complex spatial relationships. The system is designed to handle multiple imaging modalities, allowing it to process diverse datasets and support a wide range of clinical applications. More details on different model architectures developed within this project could be found in [7]–[12].

The extracted features are then transformed into structured representations that serve as input to the language model component. This transformation involves encoding spatial, semantic, and contextual information into a format that can be effectively interpreted by large language models. The chatbot module utilizes these representations to generate comprehensive radiology reports that mimic the style and content of

expert clinicians (Fig. 3). In addition to report generation, the chatbot supports interactive communication, enabling users to ask questions, request clarifications, and explore alternative interpretations. More details could be found in [13], [14].

A central component of the proposed framework is the integration of a human-in-the-loop validation mechanism. Expert radiologists are actively engaged in the review and refinement of generated outputs, thereby ensuring high levels of accuracy, clinical relevance, and reliability. This iterative process also facilitates the collection of high-quality feedback, which is leveraged to support continuous model improvement and adaptive learning. In parallel, the framework incorporates dedicated safeguards to address privacy and security concerns, particularly in scenarios involving interactions with external language models. Building upon these principles, we introduce a multi-agent reinforcement learning framework that functions as both a benchmark and an evaluation platform for multimodal clinical reasoning within the radiology ecosystem (Fig. 4). The framework integrates LLMs and VLMs within a modular architecture comprising ten specialized agents, each responsible for distinct tasks such as image analysis, feature extraction, report generation, review, and evaluation. This modular design enables fine-grained assessment at both the individual agent level and the system level, while maintaining a human-in-the-loop paradigm to ensure robust and clinically aligned performance.

## VI. CURRENT STATUS AND RESULTS

Recent experimental studies on VLMs for radiology reporting provide important insights into the design of interactive diagnostic systems. These studies demonstrate that model performance is highly dependent on architectural choices, data composition, and training strategies. While larger models often achieve higher scores in certain evaluation metrics, they also require significantly greater computational resources and do not always yield proportional improvements in clinical relevance. One notable observation is that existing evaluation metrics, such as BLEU and ROUGE, are insufficient for capturing the clinical correctness of generated reports [15]. These metrics are sensitive to syntactic similarity but may fail to reflect meaningful differences in medical interpretation. As a result, the use of reasoning-based evaluation models and expert human assessment has become increasingly important. These findings highlight the need for systems that prioritize interpretability, robustness, and human oversight.

The proposed framework incorporates these insights by emphasizing interactive reasoning and validation. Instead of relying solely on automated outputs, the system allows users to engage with the AI, explore its reasoning process, and provide feedback. This approach enhances transparency and trust, which are essential for clinical adoption. Additionally, the system is designed to balance performance and computational efficiency, enabling practical deployment in real-world environments. We reported evaluation results for benchmarking several open-source VLMs in brain cancer diagnosis using multisequence MRI [13]. A summary of these results is shown

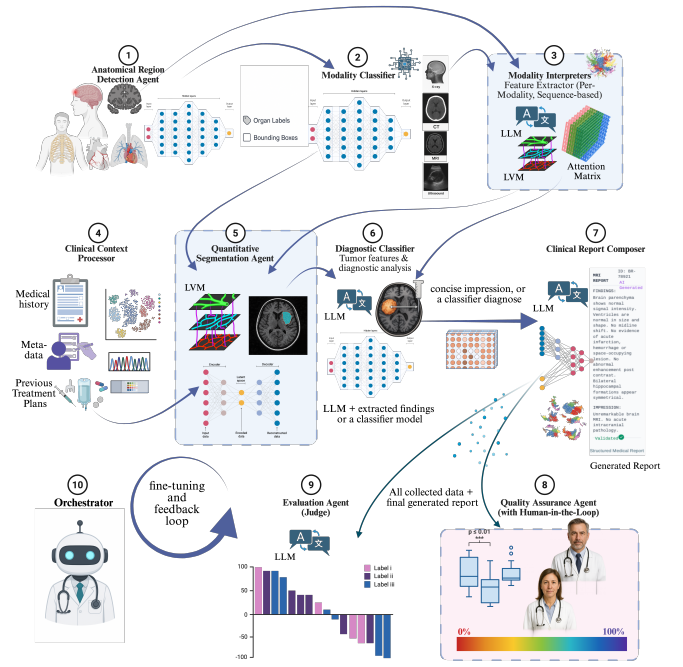


Fig. 4. Medical AI Consensus: A Multi-Agent Framework for brain tumor diagnosis [14].

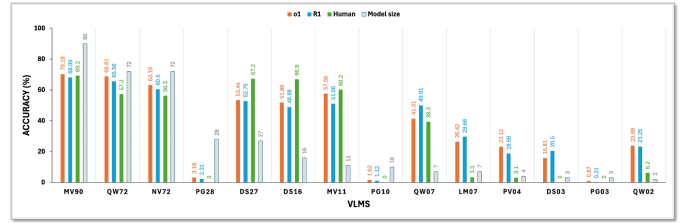


Fig. 5. Accuracy of radiology report generation for brain cancer patient using multimodal MRI using different VLMs. Evaluation is conducted using ChatGPT o1 and DeepSeek R1 (LLM as judge) and 3 human experts [13].

in Fig. 5. The findings demonstrate that the Meta-Vision 3.2 90B model achieved the highest performance. Meanwhile, mid-scale models such as Meta-Vision 11B and DeepSeek-VL-2-27B, particularly when employing iterative inference strategies, delivered an effective trade-off between accuracy and computational efficiency, performing competitively with larger models. Among the smaller-scale models, Qwen2-VL-2B showed the strongest results. Overall, these observations suggest that model scale alone does not dictate performance; instead, careful model selection and the choice of inference strategy play a critical role. The analysis further revealed existing limitations of current vision-language models in fine-grained zero-shot medical image interpretation, while also demonstrating that iterative approaches can substantially improve report generation quality, with DeepSeek-VL-2-27B exhibiting notable gains in descriptive richness under this setting.

In a typical scenario, a medical image such as a multisequence MRI scan is provided as input to the system. The image

analysis module processes the data and generates a structured representation of detected abnormalities and relevant features. This information is then used to produce a detailed radiology report (see, Fig. 6). Users can interact with the system by asking questions related to the diagnosis, requesting explanations of specific findings, or exploring alternative scenarios. The system responds in natural language, providing context-aware answers based on the extracted image features. This interaction demonstrates the system’s ability to bridge the gap between automated analysis and human understanding, offering a more intuitive and accessible approach to medical AI. The demonstration also highlights potential applications in clinical practice, medical education, and patient communication. For example, the system can be used to support junior clinicians in understanding complex cases, assist radiologists in generating reports more efficiently, and provide patients with clear explanations of their medical conditions. These capabilities underscore the versatility and impact of the proposed framework.

## VII. DISCUSSION AND FUTURE WORK

The integration of multimodal AI and conversational interfaces represents a significant step toward more human-centered healthcare systems. By enabling interactive communication and explainable reasoning, the proposed framework addresses key limitations of existing approaches and enhances the usability of AI in clinical settings. However, several challenges remain, including the need for large-scale validation, domain-specific adaptation, and robust handling of edge cases.

Ethical considerations are also critical, particularly in relation to data privacy, bias, and accountability. The system must be designed to ensure that patient information is protected and that generated outputs are fair and unbiased. Additionally, the role of human oversight must be clearly defined to prevent overreliance on automated systems. Future work will focus on addressing these challenges and further refining the framework for real-world deployment.

## VIII. CONCLUSION

This work presents an ongoing JST PRESTO research project developing an interactive multimodal AI system for medical image diagnosis, emphasizing human-centered interaction and clinical applicability. This framework builds on recent advances in vision-language models while addressing their limitations through interactive design and human-in-the-loop validation. The system has the potential to improve clinical efficiency, support medical education, and enhance patient engagement, contributing to the development of scalable and trustworthy healthcare solutions.

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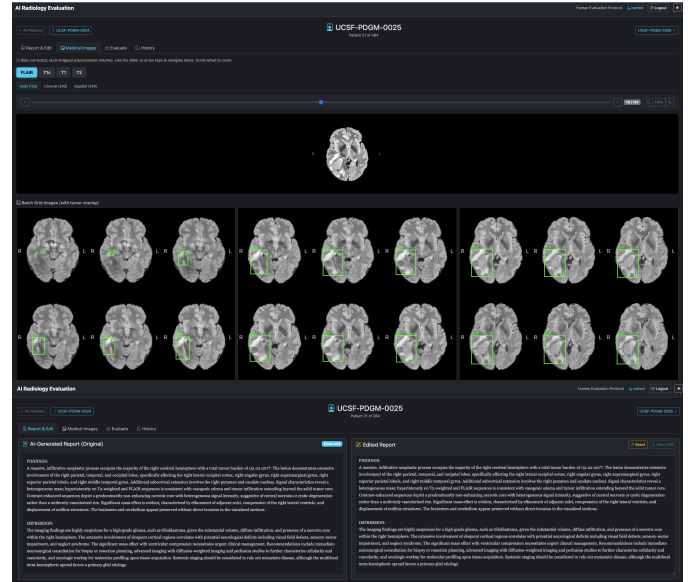


Fig. 6. A multi-agent framework for radiology report generation and evaluation. This system introduces a user experience based on real-world data. The top panel shows multi-sequence MRI images with abnormality annotations. The lower panel shows the AI-generated radiology report on the left and the report-editing window on the right. Additional evaluation tools are also available. A trial version, available in view-only mode, is accessible online [16].

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